

CONTACT

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TECHNICAL SKILLS

Languages
Python, TypeScript, JavaScript, Java, C++, SQL

Frontend
React, Next.js, Tailwind CSS, GSAP, Three.js / React Three Fiber, Zustand

Backend
FastAPI, Node.js, Express, REST APIs

Databases
PostgreSQL / PostGIS, Supabase, MongoDB, Qdrant, Oracle SQL

AI / ML
LangGraph, LangChain, Gemini API, RAG, XGBoost, ONNX, computer vision, sensor forecasting, model calibration

Data
Power BI, DAX, Pandas, NumPy, GIS

Tools & Delivery
Git, GitHub Actions, Docker Compose, Vercel, Netlify, Render, pytest, Playwright

CERTIFICATIONS

- Oracle AI Database SQL Certified Associate
  Oracle University, Aug 2026
- Geographic Information Systems, Elite (68%)
  NPTEL / IIT Roorkee, Jan-Apr 2026
- Introduction to Deep Learning
  Infosys Springboard, Jul 2026
- Java Foundation; Programming using Java; Basics of Python
  Infosys Springboard, 2025
- Software Engineering and Agile Development
  Infosys Springboard, Dec 2024
- AI for Beginners
  HP LIFE, Jul 2025
- Power BI Workshop
  OfficeMaster, Aug 2025

LOKTRISHAL K

AI & Full Stack Developer | B.Tech AI & Data Science

CAREER OBJECTIVE

B.Tech AI & Data Science student (CGPA 8.0) who builds agent workflows, ML pipelines and the web apps around them. Recent work: farm agents behind a safety gate, a seven-language healthcare navigator and an exoplanet classifier that runs in the browser. Looking for an AI or full-stack internship.

EDUCATION

B.Tech | Artificial Intelligence & Data Science — Expected Jan 2028
R.M.K. College of Engineering and Technology, Tamil Nadu | CGPA: 8.0

AWARDS & HACKATHONS

- **Innovation Award**, Cyber AI Hackathon 2026 (University of Derby, UK; supported by AMSE), Oct 2026
- **Winner**, PrototypeX, IES Summit 2026 (Jeppiaar Engineering College), Sep 2026
- Participant: ISRO Bhartiya Antariksh Hackathon 2026, HackWithHyderabad 3.0, DECODE SIH 2026, DEV HOURS Codeathon (IEEE SB)

PROJECTS

Solarda: Agentic Smart Agriculture
Agentic AI Hackathon 2026

React, TypeScript, Express, FastAPI, LangGraph, ONNX Runtime, AquaCrop-OSPy, Gemini, Docker

- LangGraph farm workflow where agents only propose actions. A deterministic TypeScript safety engine decides what runs, risky actions wait for a human, the gate fails closed if the API is down, and every action is logged.
- Plant-disease classifier (EfficientNetV2-B0, ONNX) that abstains when unsure: 98.2% on PlantVillage, 61.0% on PlantDoc field photos, and 87% correct on the images it chooses to answer.
- Crop and soil-water twin (Chennai 2021-23 weather) used about 19-32% less water than a rule-based controller at similar yield. Results come from replayed data and simulation, not live sensors.

KinSight: Inherited-Risk Screening Navigator
Future Health Digital Innovation Challenge

Next.js, TypeScript, FastAPI, LangGraph, PostgreSQL, Redis, Docker

- Explains screening results in plain language in seven languages (English, Arabic, Hindi, Tamil, Telugu, Kannada, Malayalam), with right-to-left layout for Arabic.
- A deterministic genetics engine produces every number. The couple view opens only after both partners agree (higher-risk cases also need a counsellor). Tamper-evident audit log; all data is synthetic.

ExoDetect: Exoplanet Candidate Detection
ISRO Bhartiya Antariksh Hackathon 2026

Python, XGBoost, ONNX Runtime Web, FastAPI, lightkurve, NASA MAST

- Pipeline from raw Kepler light curves (Savitzky-Golay detrending, BLS period search, 13 transit features) to a star-level 70/15/15 split, so one star never lands in two splits.
- XGBoost with isotonic calibration, a threshold picked on validation only, and TreeSHAP explanations. Exported to ONNX so the classifier runs in the browser. On a 400-star sample, AUC-PR fell to 0.913 once depth and SNR features were removed; a harder benchmark is still pending.

NWIS: Nearby Wells Intelligence System
Hackathon prototype

React, FastAPI, PostGIS, Qdrant, LangGraph, Docker

- Ranks offset wells by similarity instead of distance alone and ties each risk estimate to a source report and page. A four-agent LangGraph flow answers with citations or says the evidence is insufficient.
- Role-based access and a hash-chained audit log. 200 backend and 51 frontend tests pass on a synthetic 12-well dataset, so none of it reflects real field data.

Legal & Regulatory Assurance Platform
In development

FastAPI, React, PostgreSQL, Qdrant, PyMuPDF

- Secure document intake (hashing, quarantine), TXT/DOCX/PDF extraction with page-level locators, and a resolver that returns exact stored quotes. 126 scope and security regression tests pass; contract and regulatory analysis are still planned.

OTHER PROJECTS

- **ResumeForge AI** *(Next.js, LangGraph, Gemini 2.0, Supabase)*: six-agent resume and career pipeline with Google/GitHub OAuth and OTP sign-in.
- **SentinelPay** *(React, Node.js, MongoDB, Gemini API)*: UPI fraud risk scoring, deployed on Vercel with GitHub CI/CD.
- **PeopleRamp / OnboardIQ** *(React, Node.js, Gemini)*: AI onboarding tools with voice, text and camera input and a 30-60-90 day plan.
- **EduNova** *(Next.js 15, React Three Fiber)*: science-learning site with three video-quest worlds and parent/teacher views.
- **Sri Ambal Healthcare** *(HTML/JS, Supabase, Gemini)*: hospital site with booking saved to Supabase and Google Sheets, plus a chatbot.
- **Amazon Sales Dashboard** *(Power BI, DAX, SQL)*: cleaned raw sales data; built KPI, trend and category views.